

Artificial Intelligence

AI-2002

Lab 02 Tasks

OOP in Python

National University of Computer & Emerging Sciences –
NUCES – Karachi

Department of Software Engineering



LAB TASKS

TASK #1

You are developing a simple AI-powered healthcare monitoring system to manage different types of healthcare agents. All agents share common attributes such as `agent_id`, `patient_name`, and `status`, but each type has its own specialized role. For example, `HeartMonitoringAgent` monitors a patient's heart rate and vital signs, `MedicineReminderAgent` reminds patients to take their prescribed medicines on time, and `HealthPredictionAgent` analyzes patient data to identify possible health risks. Your task is to create a base class `HealthcareAgent` and derive classes for each agent type, implementing type-specific methods such as `monitor_heart_rate()`, `remind_medicine()`, or `predict_health_risk()`. Create objects for each agent type and call their methods to simulate how an AI-powered healthcare system can continuously monitor patients, provide reminders, and assist in early health-risk detection.

TASK #2

You are building an AI-based smart agriculture monitoring system that manages different types of agricultural threats. All threats share common attributes such as `threat_id`, `name`, and `severity`. Derived classes include `CropDiseaseThreat` (monitors crops for signs of disease), `PestThreat` (monitors crops for harmful insects and pests), and `WaterStressThreat` (monitors soil moisture and detects insufficient irrigation). Each class has specific methods such as `detect_disease()`, `detect_pests()`, or `check_soil_moisture()`. Your task is to create the base and derived classes, instantiate objects for each threat type, and call their methods to simulate an automated AI-based agricultural threat detection and response system.

TASK #3

You are designing an AI system to automatically respond to cybersecurity threats. There are different types of response agents: `AlertAgent`, `BlockAgent`, and `RecoverAgent`. All agents have a method `execute_response()`, but each performs a different action—`AlertAgent` sends notifications, `BlockAgent` blocks malicious activities, and `RecoverAgent` restores affected systems. Your task is to create a base class `ResponseAgent` with a generic `execute_response()` method, override it in each derived class, and then demonstrate polymorphism by calling `execute_response()` on a list of mixed agent objects.

TASK #4

You are developing a simple employee management system. The base class `Employee` has a method `work()`. Derived classes like `Manager`, `Developer`, and `Designer` override `work()` with tasks specific to their role. Students should create objects of each employee type and call `work()` to show how polymorphism allows the same method name to perform different actions.



National University of Computer & Emerging Sciences - NUCES - Karachi
Department of Software Engineering

TASK #5

You are creating a UAV flight management system. Each UAV has a private attribute `__battery_level` representing its current battery percentage. The class `UAV` should have methods `set_battery_level(level)` to update the battery level, `get_battery_level()` to retrieve the current battery level, and `display_info()` to show the UAV's ID, model, and battery level. The `set_battery_level()` method should ensure that the battery level remains between 0 and 100. Students should create multiple `UAV` objects, update their battery levels using the setter method, retrieve battery levels using the getter method, and understand how encapsulation protects important UAV data from direct modification.